



Exercises 3 : Interaction and Concurrency

Luís Soares Barbosa

Exercise I.1

Let $A(a) \triangleq a.A$ and $B(b) \triangleq \bar{b}.B$. Compute the first derivatives of the following processes:

1. $A + B$
2. $A + B\langle a \rangle$
3. $A \mid B$
4. $A \mid B\langle a \rangle$
5. $(A \mid B)[a/b]$
6. $(A \mid B\langle a \rangle) \setminus \{a\}$

Exercise I.2

Let $A(a, b, c, d) \triangleq \bar{a}.b.A + \bar{c}.d.A$. Draw the transition graphs of the following processes

1. A
2. $A \setminus \{a\}$

Exercise I.3

Consider the following description of a two-position *buffer* with acknowledgements. Note the process is built from copies of a 1-position *buffer* also with acknowledgements: it acknowledges in $\bar{r}$ the reception of a message and waits in t the confirmation that a message sent was arrived to its destination.

$$Bs \triangleq (B(in, mo, mi, r) \mid B(mo, out, t, mi)) \setminus \{mo, mi\}$$
$$B(in, out, t, r) \triangleq in.\overline{out}.t.\bar{r}.B$$

1. Draw the synchronisation graph of Bs .
2. Check whether the behaviour of Bs is the intended one (drawing, for this purpose, the corresponding transition graph)
3. Find a solution to the problem detected (if any) and draw the corresponding transition graph.
4. Explain how the specification given (or your new solution) can be adapted to describe *buffers* with an arbitrary, but fixed number of positions.

Exercise I.4

Consider the following description of a 1-position bidirectional buffer, i.e. able to transmit and receive messages in any direction.

$$BT(in_1, in_2, out_1, out_2) \triangleq in_1(x).\overline{out_1}(x).BT + in_2(x).\overline{out_2}(x).BT$$

1. Specify a 2-position bidirectional buffer by parallel composition of two instances of process BT .
 2. Draw its synchronisation diagram and the transition graph.
-

Exercise I.5

An n -trigger, for $n > 1$, is used in electronic voting to detect that a fixed number of votes have been received along its n input ports, numbered from a_1 to a_n . As soon votes have been received in half of the input ports a signal is sent through its output port $\bar{s}$ and the process terminates. Each port a_i receives only a single input. Inputs, however, may arrive in any order to the different ports.

1. Specify a 3-trigger.
 2. Specify a n -trigger, for n arbitrary.
-

Exercise I.6

Draw the transition graph of $T \triangleq a.(b.0 \mid T)$?